

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript, C++, Swift, SQL

Tools and Frameworks: REST(FastAPI, Next.js), React, PostgreSQL, AWS, Cloudwatch, Splunk, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD), Git, Linux, JUnit, VS Code, XCode

Areas of Interest and Coursework: Distributed System Design, Software Engineering(SDLC), AI & LLMs

ACADEMIC EXPERIENCE / PROJECTS

Graduate Assistant | UMBC - Data Structures & Software Engineering

Jan 2025 – June 2026

- Mentored 10 student **Agile** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.
- Helped 50+ students design and debug C++ projects implementing core data structures, and graded their homework.
- Held office hours to clarify technical concepts and translate requirements into **actionable implementation steps**.
- Designed over 10 C++ assignments on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.

Role Radar | Desktop app for the latest job-posting alerts

- Designed **native desktop apps** in SwiftUI (macOS) and C#/.NET (Windows) with a guided setup and live job-match tracking that close to the menu bar or tray and supervise the checker daemon (launchd on macOS, a Run-key login process on Windows), so non-technical users get alerts with **no terminal or config files**.
- Built an async Python job-ingestion engine that tracks 7,700+ employers' careers pages, auto-selecting 1 of 22 source-specific scrapers (Greenhouse, Workday, Oracle HCM, ...) by URL and routing every request through a robots.txt-compliant client with per-host rate limits and Retry-After backoff, checking ~3,600 companies per 18-minute pass across ~1,000 hosts at a **0.18%** request failure rate.
- Designed a **laptop-to-Lambda failover** that keeps checks running while the Mac sleeps, electing a single runner through a conditionally written DynamoDB lease whose epoch acts as a **fencing token** on each company's transactional save, so a laptop waking mid-run can never overwrite its successor's state, with Lambda taking over within **8 min** and handing back in **~15 s**.

INDUSTRY EXPERIENCE

Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Containerized the platform (FastAPI, Postgres, NGINX) and deployed it to **AWS EKS** with Kubernetes manifests and a GitHub Actions **CI/CD pipeline**, adding health checks and admin endpoints to download models and rebuild search indexes, so the team could manage models and indexes without shelling into pods.
- Developed **backend services** for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.
- Built a priority-ordered fallback in the **RAG chat agent** across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment accuracy from **64% to 92%**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an AI-powered PRD review platform in React and FastAPI that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully **serverless** review pipeline on **AWS** (S3 + CloudFront-hosted React SPA, FastAPI on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining 200+ submissions across 5 cohorts at **~\$3/month** idle cost.

- Architected a self-serve quoting platform that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from **~2 days** to **<1 min**.
- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to **Kafka** workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across **~1,100** orders.
- Delivered the **React SPA** that carried a customer from property details to instant estimate to paid deposit in one client-routed flow, with role-gated admin routes and JWT access tokens backed by httpOnly refresh cookies, enabling a streamlined booking flow with **~40%** of bookings occurring outside office hours without phone support.
- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for **~10×** seasonal peaks and load-tested to **~50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.